

VEHICLES' ETHERNET

Emerging Trends And Challenges



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Connected car is the buzz word today. In reality, the connected car already relies on the Ethernet. From infotainment systems and backup camera connections, the Ethernet's high capacity and versatility readily penetrates automotive segments.

Ethernet migration in the automotive market in 2016 is a new trend. The advanced driver assistance system featured in many vehicles is expected to expand the use of the Ethernet. Recent technological developments make the Ethernet viable for use in cars. As per estimates by Frost & Sullivan and Strategy Analysis, nearly 400 million automotive Ethernet ports will be in use by 2020. By then, total number of automotive Ethernet ports is expected to be higher than the total number of all other Ethernet ports combined.

Gigabit Ethernet is also in the pipeline in 2016. Emergence of 1000BASE-T1 standard in mid-2016 provides a roadmap for the automotive Ethernet evolution. By 2020, it is estimated that 40 per cent of the cost of a car will be in electronics.

As automotive manufacturers are adding more and more computer based

systems, applications and connections, cost of electronics content and the wiring harness to support these in terms of cabling, network interfaces and onboard computing power is growing. Ethernet deployment can, and will, reduce these costs.

Media-oriented systems transport (MOST) currently dominates as a dedicated high-speed multimedia network technology for auto infotainment systems. MOST bus, based on a ring topology, uses synchronous data communication to transport audio, video and voice signals. With Gigabit Ethernet, it could support many more streams than MOST at a higher data rate in higher resolution.

A vehicle today consists of an infotainment, advanced driver assistance system (ADAS) and control domains, each using MOST, Flexray, CAN, LIN, LVDS and the Ethernet. Even though the Ethernet has existed for over 20 years, it could not be previously used in automobiles due to the drawbacks like it did not meet OEM EMI/RFI requirements for the automotive market, was susceptible to alien noise from other devices in a car, could not guarantee latency down to the low microsecond range and did not have a way of synchronising time between devices and having multiple devices sample data at the same time.

Automotive Ethernet

Automotive Ethernet is a physical network that is used to connect components within a car using a wired network. It is designed to meet the needs of the automotive market, including meeting electrical requirements such as EMI/RFI emissions and susceptibility, bandwidth requirements, latency requirements, synchronisation and network management requirements.

Fig. 1: Electronics anatomy using automotive Ethernet

